Eric Nieto Gonzalez

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Education:

University of Illinois at Urbana-Champaign — Grainger College of Engineering

- M.S. in Electrical Engineering — Expected Graduation: May 2027
  - GPA: 3.93 / 4.0
- B.S. in Electrical Engineering — August 2020 → May 2025
  - GPA: 3.55 / 4.0

Work Experience:

Yield Excellence Engineer Intern at TSMC — May 2026 → August 2026

- Optimized SEMV G10 recipe parameters for N4 FinFET to increase inspection throughput by 20%
- Hands-on experience with semiconductor software such as KLARITY, Lot Action, and OMI
- Reviewed 50 SEM images per day to classify wafer defects and perform root cause analysis

Teaching Assistant for ECE 340: Semiconductor Devices — August 2025 → Present

- Created 1 original problem per assignment to reinforce semiconductor concepts
- Facilitated weekly office hours, providing individual support to 10 students per session
- Evaluated student performance by grading homework and exams through constructive feedback

Manufacturing Engineering Intern at G&W Electric Co. — May 2024 → August 2024

- Designed custom tools that improved manufacturing efficiency by 40%
- Proficient in industry engineering documentation, including product and electrical schematics
- Coordinated with 12 employees across all specialties to collect input for project development

Publications:

- **Eric Nieto Gonzalez**, S. Edward, M. Gallegos, T. Golecki, B. Kaehr, & H. M. Golecki, "Multiphoton-Printed Microrobotic Arrays Powered by Discrete, Monolithically Integrated Hydrogel Actuators," *Advanced Robotics Research*, 2026, doi:10.1002/adrr.202500186

Projects:

Nanoscale Semiconductor Devices and Simulation — January 2025 → May 2025

- Studied nanoscale semiconductor properties and carbon nanomaterials
- Gained hands-on experience with nanofabrication techniques within a cleanroom
- Evaluated surface characteristics through an SEM

Advanced Optoelectronic Device Characterization — January 2025 → May 2025

- Explored energy conversion devices for LEDs and solar cells
- Analyzed heterostructures and low-dimensional quantum structures for modern devices
- Characterized LEDs and solar cells using TCAD simulations for electrical and optical modeling

Integrated Circuit Wafer Fabrication and Testing — August 2024 → December 2024

- Fabricated semiconductor devices using photolithography, diffusion, and etching techniques
- Developed cleanroom skills to produce functioning MOS transistors and diodes
- Conducted device testing to ensure functionality, identify defects, and validate the design.
- Analyzed process parameters to refine microfabrication techniques and improve device yield.

Developed SNES Street Fighter 2 using System Verilog — April, 2023 → May, 2023

- Coded modules in System Verilog which handle multiple sections of the game simultaneously
- Numerous modules created to make it more efficient to debug, understand, and improve
- Drivers integrated to receive keyboard signals and send out VGA signals within the board
- Wrote a thorough, concise, and formal lab report relating to the specific details of the game

Skills:

- PCB Design, KiCad
- System Verilog, HDL
- C Language
- MATLAB
- PTC Creo / SolidWorks
- Seal of Biliteracy in Spanish & English

Awards:

- UIUC Knight of St. Patrick's Awardee
- Intel Scholar
- ILLCF Scholarship Recipient
- Hispanic Scholarship Fund Recipient
- HRP Scholarship Recipient
- Wentcher Scholarship Recipient